

本次作業使用 SystemC 實作一個 4×4 mesh-based NoC。整體架構包含 16 個 routers 和 16 個 cores。每個 core 透過 Network Interface 將 PE 產生的 packet 拆成 34-bit filts，並透過 router 傳送到目的地。最後目的端 core 將 filts 組回 packet 並交給 PE 檢查。

System Architecture :

本設計採用 4×4 mesh topology。每個 router 最多有五個方向：^{local} North、South、East、West、Local part 連接 core，其餘方向連接相鄰 router。Router ID 和 Core ID 從左上角依序編號 0 到 15。

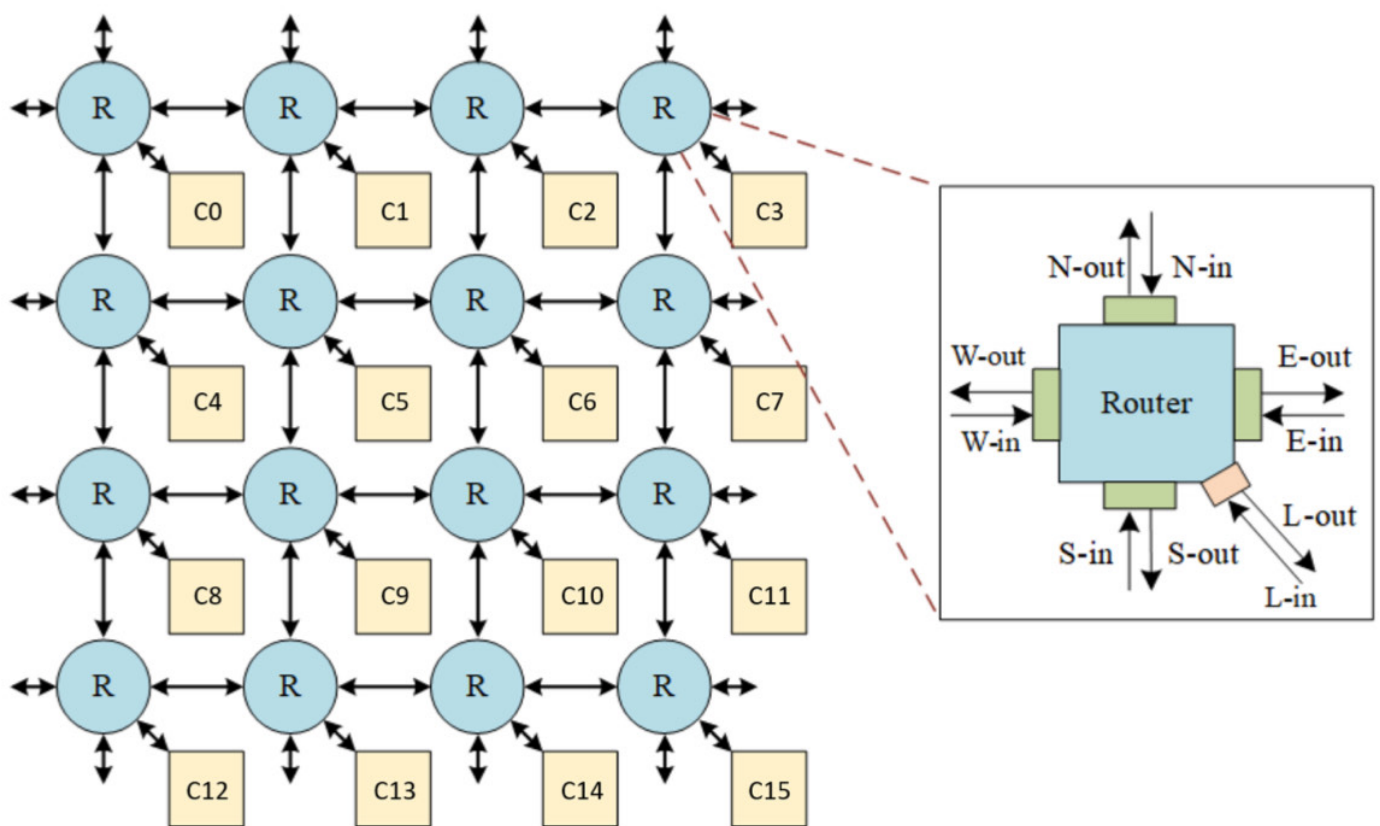


Figure 1. 4×4 mesh-based NoC architecture

Core / NI Design :

Core 內部包含 PE 和 NI。PE 不能改，NI 負責兩件事：

1. transmit path: 將 packet 拆成 header filt、body filt、tail filt
2. receive path: 將收到的 filts 組回 packet，並在收到 tail filt 後呼叫

pe_check_packet()
(filt 大小是 34 bits)

Router Design :

Router

Router 使用五組 input/output ports, 分別對應 North-South-East-West-Local。每個 input port 有一個 input queue, 每個 output port 有一個 output queue。Router 收到 header filts 後會根據 destination ID 決定 output direction, 並鎖定該 packet 的路徑, 使後續 body/tail filts 走相同 output port。

Buffer

Buffer 使用 std::queue 實作, 因此在 SystemC simulation 中沒有明確限制 queue depth。每個 input port 有一個 FIFO elastic buffer。

VC

本設計沒有使用 virtual channel。每個 physical input port 只有一個 FIFO queue, 以降低實作複雜度。

Routing Algorithm :

本設計採用 deterministic XY routing。Router 先比較目前位置和目的地位置的 X 座標, 優先往 East 或 West 移動。當 X 座標相同後再根據 Y 座標往 North 或 South 移動。若目前 router 已經是目的地, 則送到 local port。

座標轉換:
$$\begin{cases} x = \text{router_id} \% 4 \\ y = \text{router_id} / 4 \end{cases}$$

Pipeline Stage and Cycle Time :

Stage	Blocks	Delay
1	Sync In + Elastic Buffer	$0.45 + 0.44 = 0.89$
2	Routing Computation	0.30
3	Arbiter	0.23
4	Crossbar switch + Sync Out	$0.23 + 0.45 = 0.68$

$\therefore \text{Cycle time} = \text{Max}(0.89, 0.30, 0.23, 0.68) = 0.89 \text{ ns}$

Optimization :

原本的 baseline router 在每次完成 flit 傳送後，都會額外等待一個 cycle。我的修改是讓 router 只有在 output queue 為空時才 wait()。如果 output queue 中還有 flit，就可以立刻繼續傳送下一個 flit，因此減少了不必要的 idle cycles。

改善了： $1021 - 819 = 202$ cycles

Simulation Result :

```
14:17 es26mlchip070@ee24[~/hw3]$ make
g++ -I. -I /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/include -L. -L /RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64 -o run *.cpp -lsystemc-2.3.3 -Wl,-rpath,/RAID2/COURSE/2026_Spring/es26mlchip/es26mlchipTA10/systemc-2.3.3/lib-linux64
./run

SystemC 2.3.3-Accellera --- Mar  3 2026 15:59:48
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ALL RIGHTS RESERVED

=====
+                                     +
+  Congratulations !!                +
+                                     +
+  Simulation completed              +
+  at 819 th cycle                   +
+                                     +
=====
+                                     +
+  / 0.0                             +
+ / ^ ^ ^ ^ ^ \                      +
+ | ^ ^ ^ ^ ^ |                      +
+ \ m m m m m /                      +
+  \ 0.0                             +
+                                     +

Info: /OSCI/SystemC: Simulation stopped by user.
14:17 es26mlchip070@ee24[~/hw3]$
```

```
Info: /OSCI/SystemC: Simulation stopped by user.
14:40 es26mlchip070@ee24[~/hw3]$ cd 09_SUBMIT
14:40 es26mlchip070@ee24[~/hw3/09_SUBMIT]$ ./00_tar
[Info] Directory created: hw3_es26mlchip070
[Info] Copied: ../clockreset.h -> hw3_es26mlchip070/clockreset_es26mlchip070.h
[Info] Copied: ../core.h -> hw3_es26mlchip070/core_es26mlchip070.h
[Info] Copied: ../core_base.h -> hw3_es26mlchip070/core_base_es26mlchip070.h
[Info] Copied: ../pe.h -> hw3_es26mlchip070/pe_es26mlchip070.h
[Info] Copied: ../router.h -> hw3_es26mlchip070/router_es26mlchip070.h
[Info] Copied: ../router_base.h -> hw3_es26mlchip070/router_base_es26mlchip070.h
[Info] Copied: ../clockreset.cpp -> hw3_es26mlchip070/clockreset_es26mlchip070.cpp
[Info] Copied: ../main.cpp -> hw3_es26mlchip070/main_es26mlchip070.cpp
[Info] Copied: ../pe.cpp -> hw3_es26mlchip070/pe_es26mlchip070.cpp
[Info] Creating tar archive: hw3_es26mlchip070.tar.gz
hw3_es26mlchip070/
hw3_es26mlchip070/clockreset_es26mlchip070.h
hw3_es26mlchip070/core_es26mlchip070.h
hw3_es26mlchip070/core_base_es26mlchip070.h
hw3_es26mlchip070/pe_es26mlchip070.h
hw3_es26mlchip070/router_es26mlchip070.h
hw3_es26mlchip070/router_base_es26mlchip070.h
hw3_es26mlchip070/clockreset_es26mlchip070.cpp
hw3_es26mlchip070/main_es26mlchip070.cpp
hw3_es26mlchip070/pe_es26mlchip070.cpp
[Success] hw3_es26mlchip070.tar.gz created successfully.
14:40 es26mlchip070@ee24[~/hw3/09_SUBMIT]$ ./01_submit
[Info] Deadline check OK ...
[Info] File check OK ...
[Info] Your file will be submitted to: TA folder
[Info] Now submit hw3_es26mlchip070.tar.gz file to system.
[Success] Copying Successfully.

=====
Submit Report
=====
Result      : has been submitted.
Submission time : 2026/05/12 14:40:41
=====
--                                     --
--  Congratulations !!                --
--                                     --
--  Submission Successful!!           --
--                                     --
-- / 0.0                             --
-- / ^ ^ ^ ^ ^ \                      --
-- | ^ ^ ^ ^ ^ |                      --
-- \ m m m m m /                      --
--  \ 0.0                             --
--                                     --

Please remember to check your submission with ./02_check !!
Please remember to check your submission with ./02_check !!
Please remember to check your submission with ./02_check !!

=====
14:40 es26mlchip070@ee24[~/hw3/09_SUBMIT]$ ./02_check
hw3_es26mlchip070.tar.gz has been downloaded!
14:40 es26mlchip070@ee24[~/hw3/09_SUBMIT]$
```